Rapid Mobile App Development with Xamarin.Forms

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You're not James

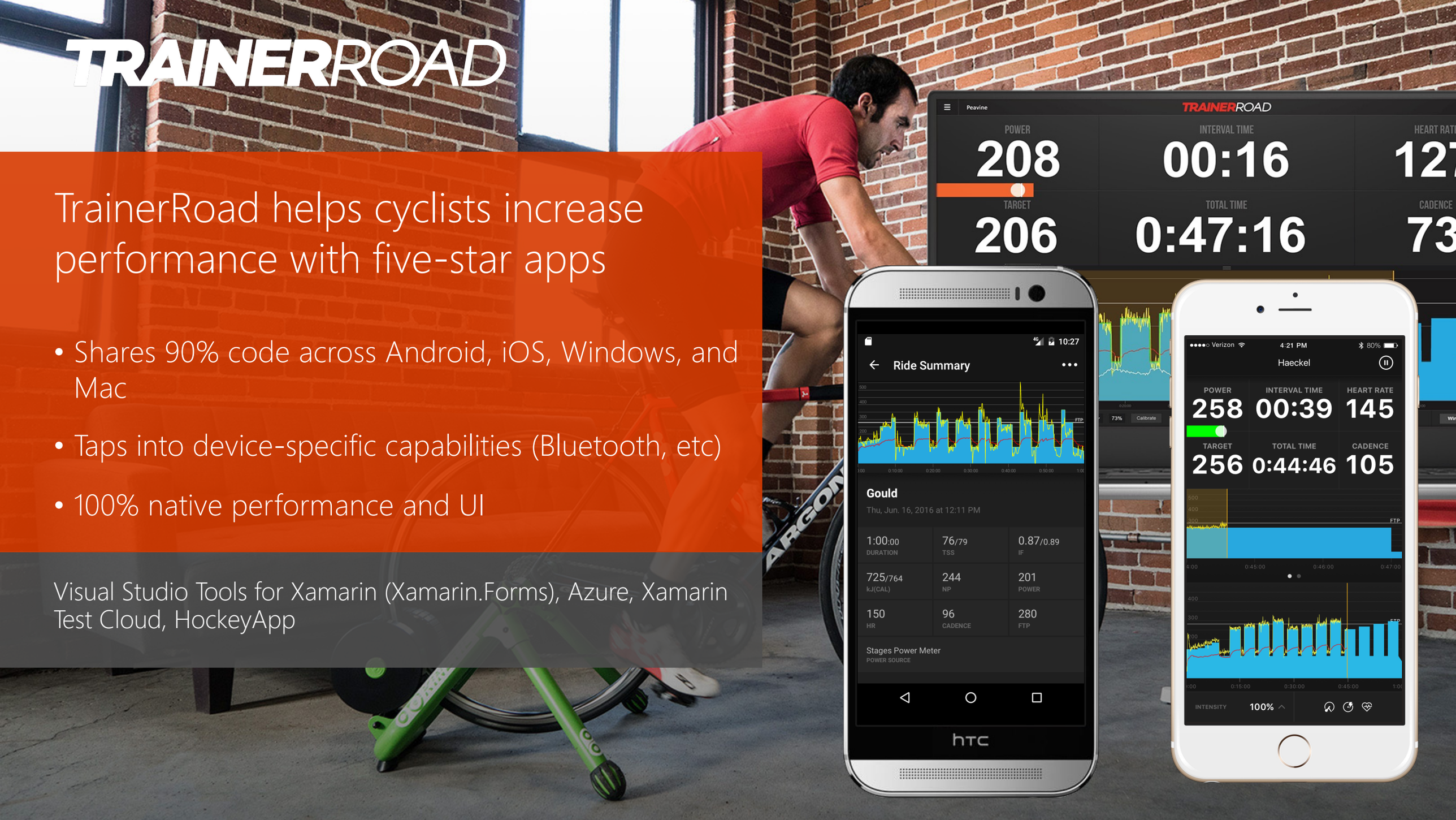


TRAINERROAD

TrainerRoad helps cyclists increase performance with five-star apps

- Shares 90% code across Android, iOS, Windows, and Mac
- Taps into device-specific capabilities (Bluetooth, etc)
- 100% native performance and UI

Visual Studio Tools for Xamarin (Xamarin.Forms), Azure, Xamarin Test Cloud, HockeyApp

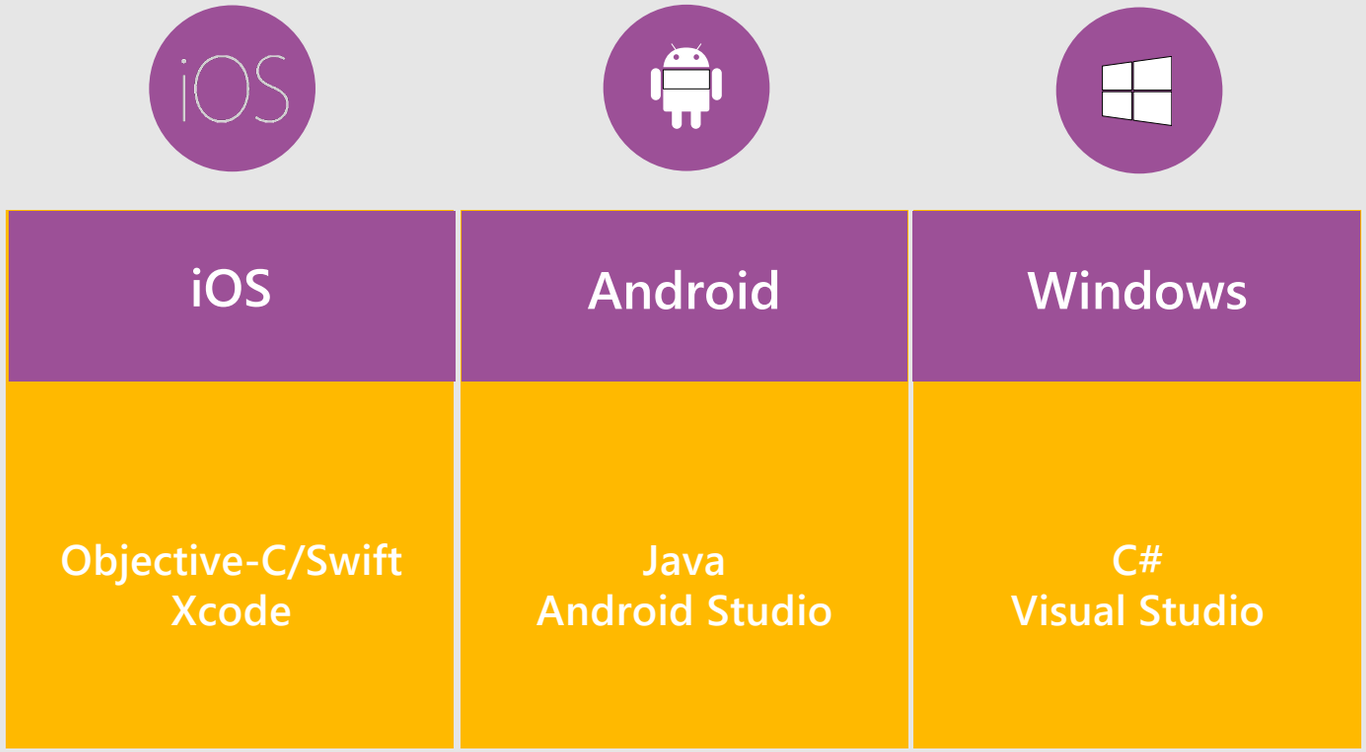


What is Native?



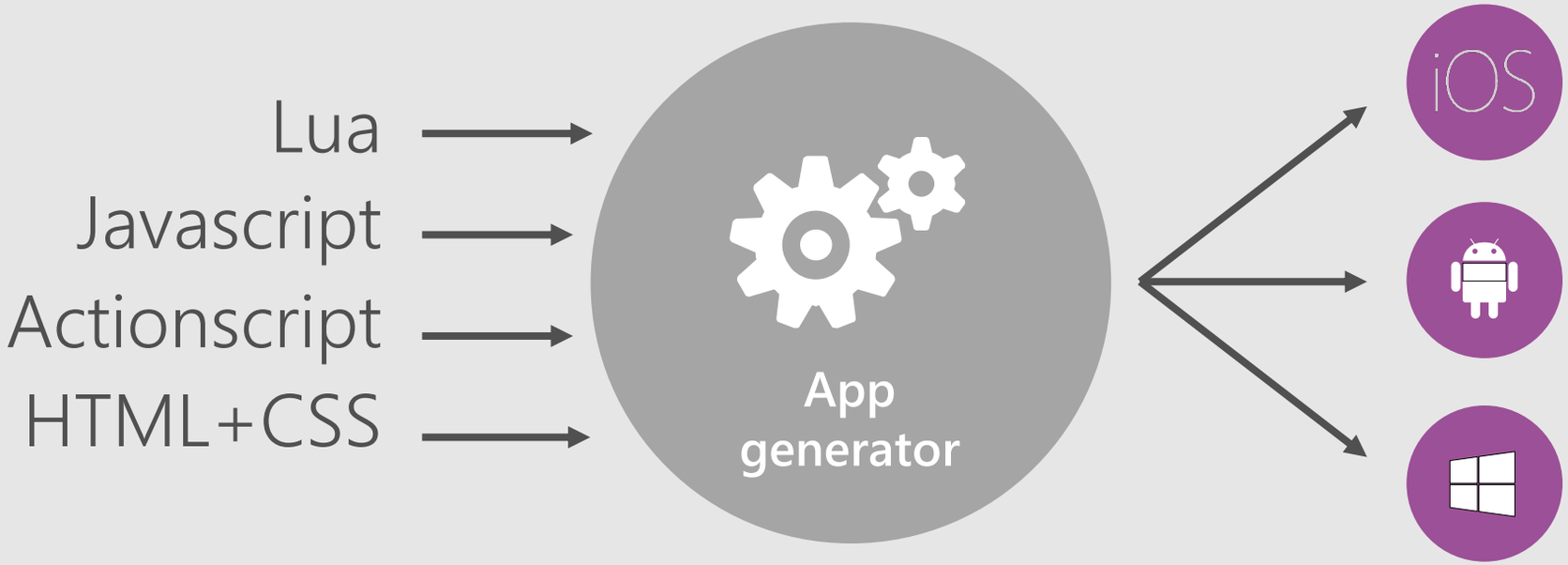
Architecting Mobile Apps

Silo approach



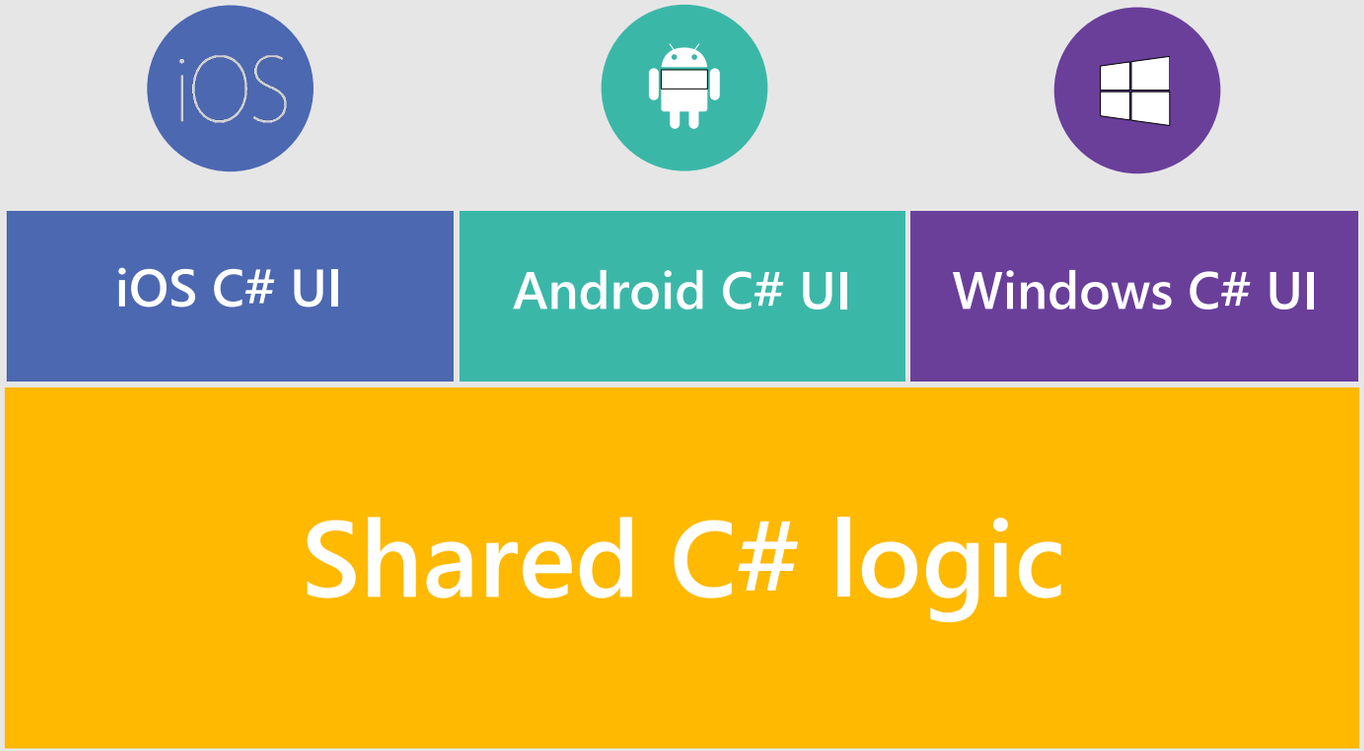
No shared code • Many languages and development environments • Multiple teams

Write once, run anywhere



Limited native API access • Slow performance • Poor user experience

Xamarin's unique approach



Shared C# codebase • 100% native API access • High performance

User Interface

Xamarin Native approach

iOS C# | Android C# | Windows C#

Shared C# Logic

3 Native User Interfaces

Shared App Logic



Xamarin.Forms - Cross Platform UI





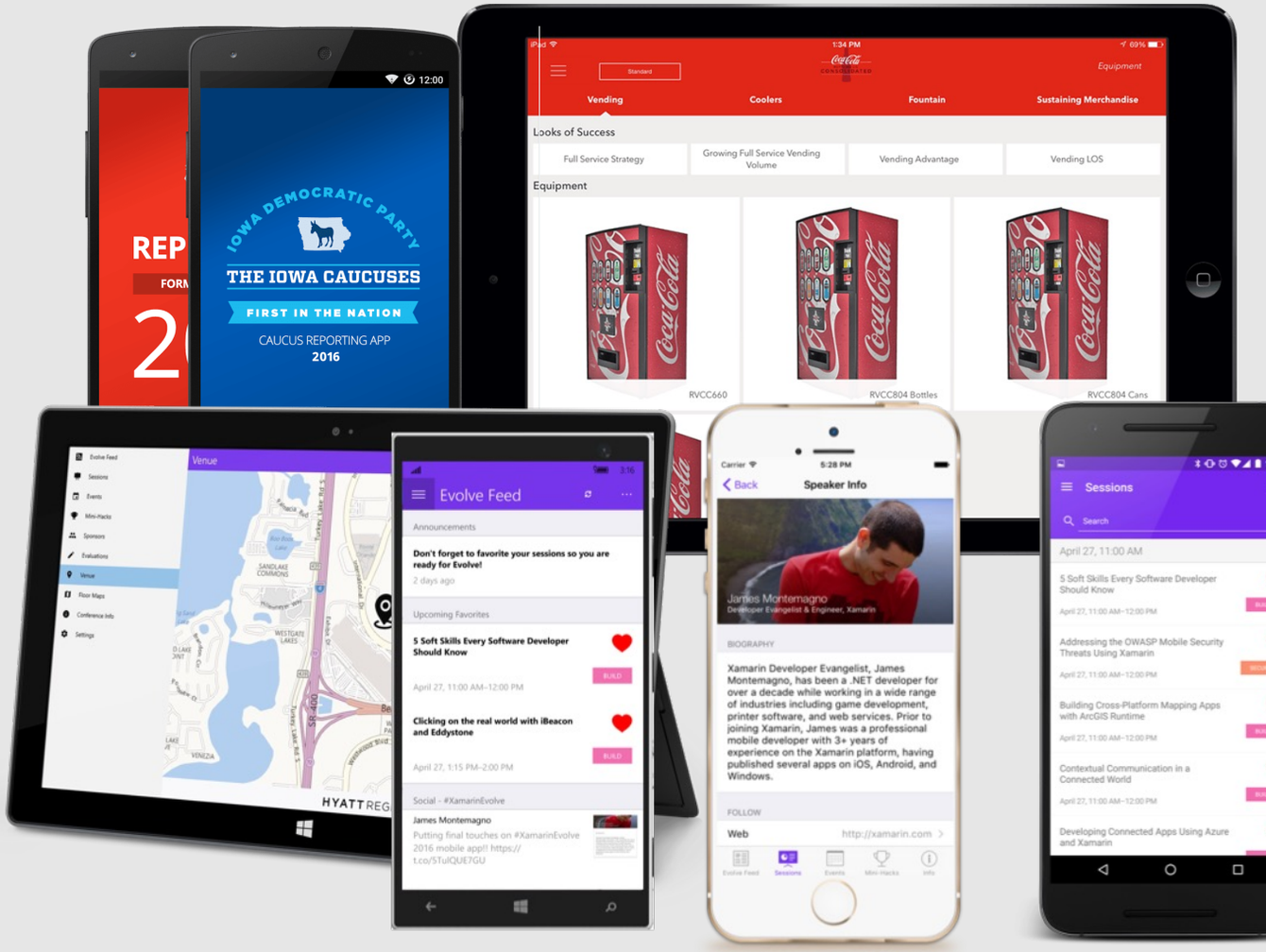


Shared UI Code

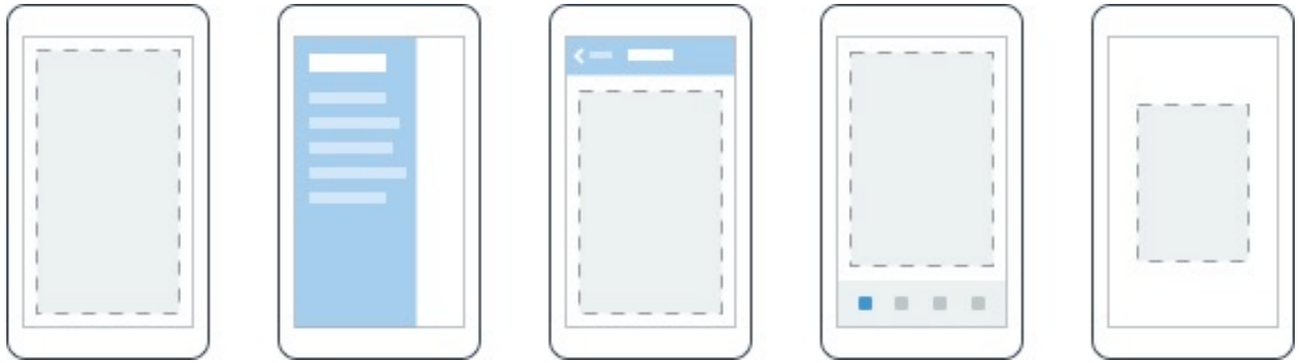
Shared C# Logic

Shared Native User Interface

Shared App Logic

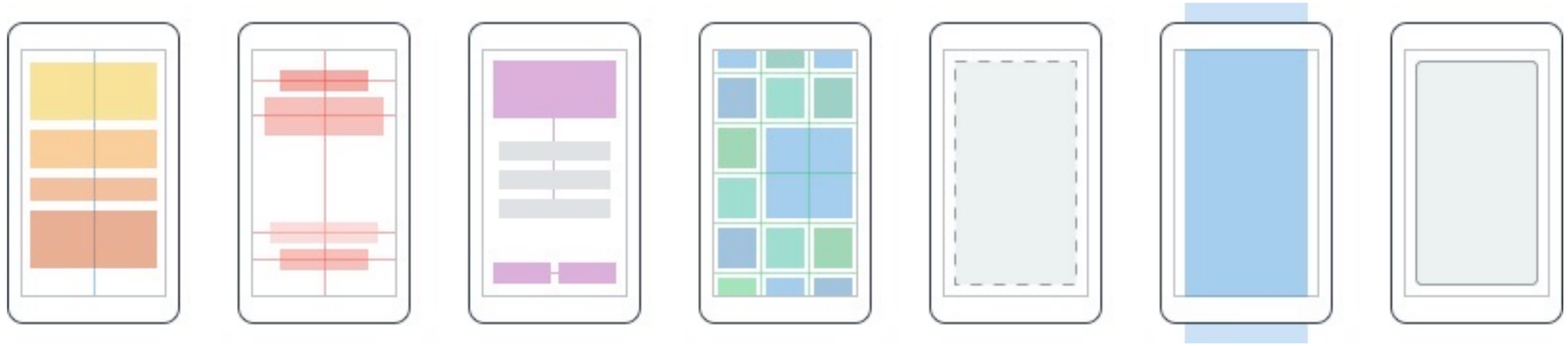


Pages



Content MasterDetail Navigation Tabbed Carousel

Layouts



Stack Absolute Relative Grid ContentView ScrollView Frame

Controls

ActivityIndicator	BoxView	Button	DatePicker	Editor
Entry	Image	Label	ListView	Map
OpenGLView	Picker	ProgressBar	SearchBar	Slider
Stepper	TableView	TimePicker	WebView	EntryCell
ImageCell	SwitchCell	TextCell	ViewCell	

Native UI

Forms elements are realized via platform renderers.

	ContentPage	Label	Picker
Android	Activity / Fragment	TextView	AlertDialog + LinearLayout + EditText + NumberPicker
iOS	UIViewController	UILabel	UITextField + UIPickerView + UIToolbar + UIBarButtonItem
UWP	Page	TextBlock	ComboBox

developer.xamarin.com/guides/xamarin-forms/application-fundamentals/custom-renderer/renderers/

Bundled Services

DependencyService

MessagingCenter

Navigation

Themes

3rd Party App Frameworks

MvvmCross

MvvmLight

Prism

FreshMvvm

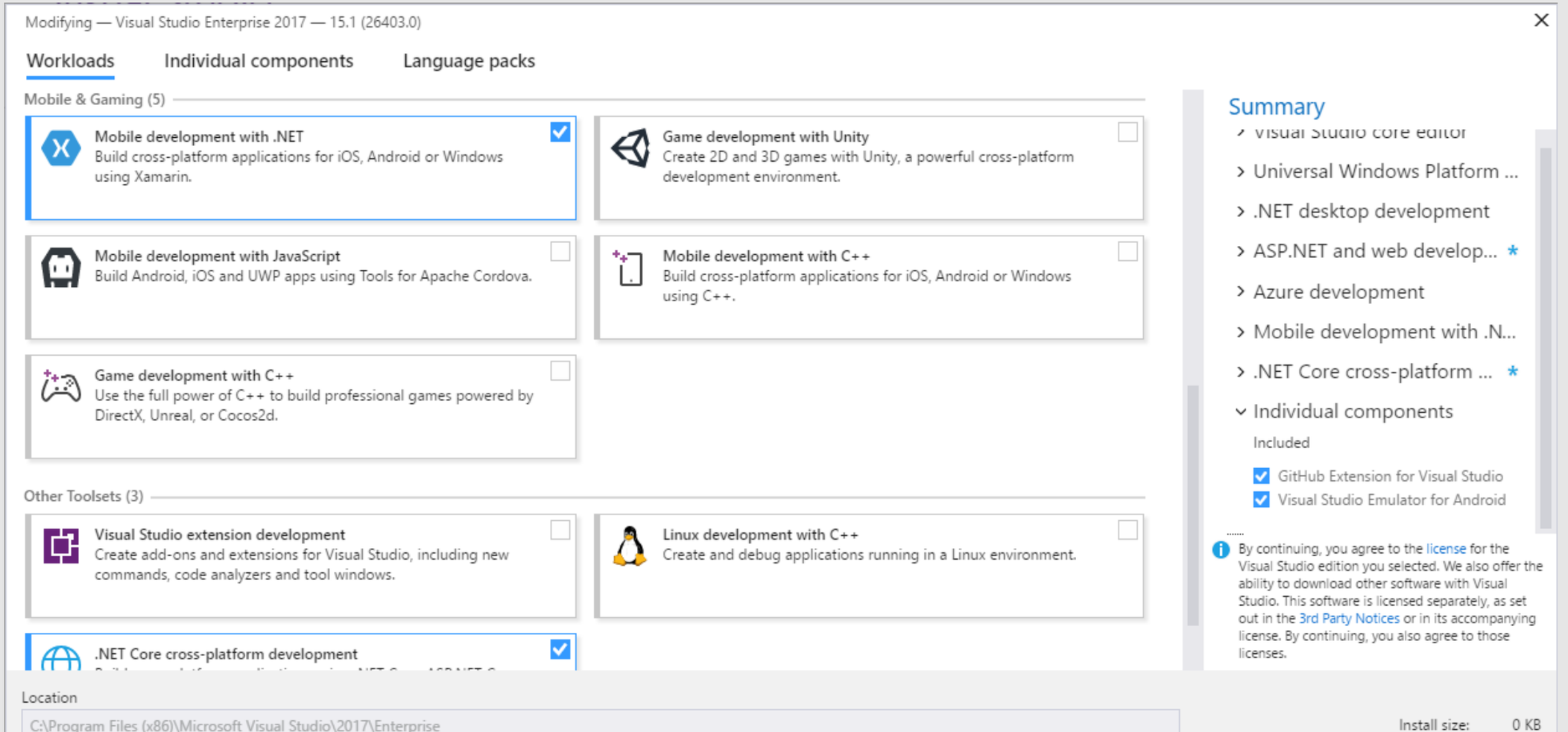
ReactiveUI

And more…

Your Development Environment

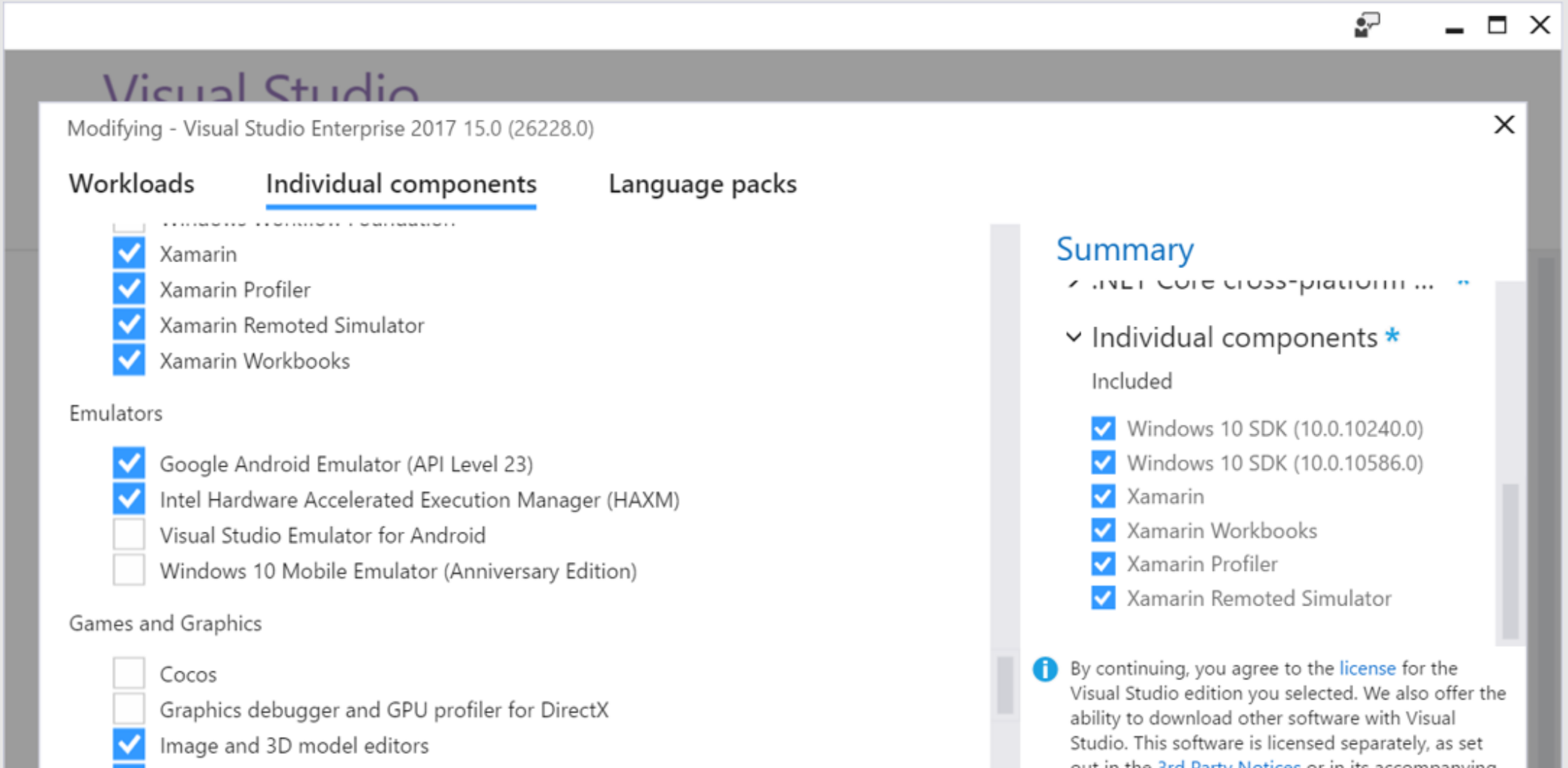
Mobile Workload

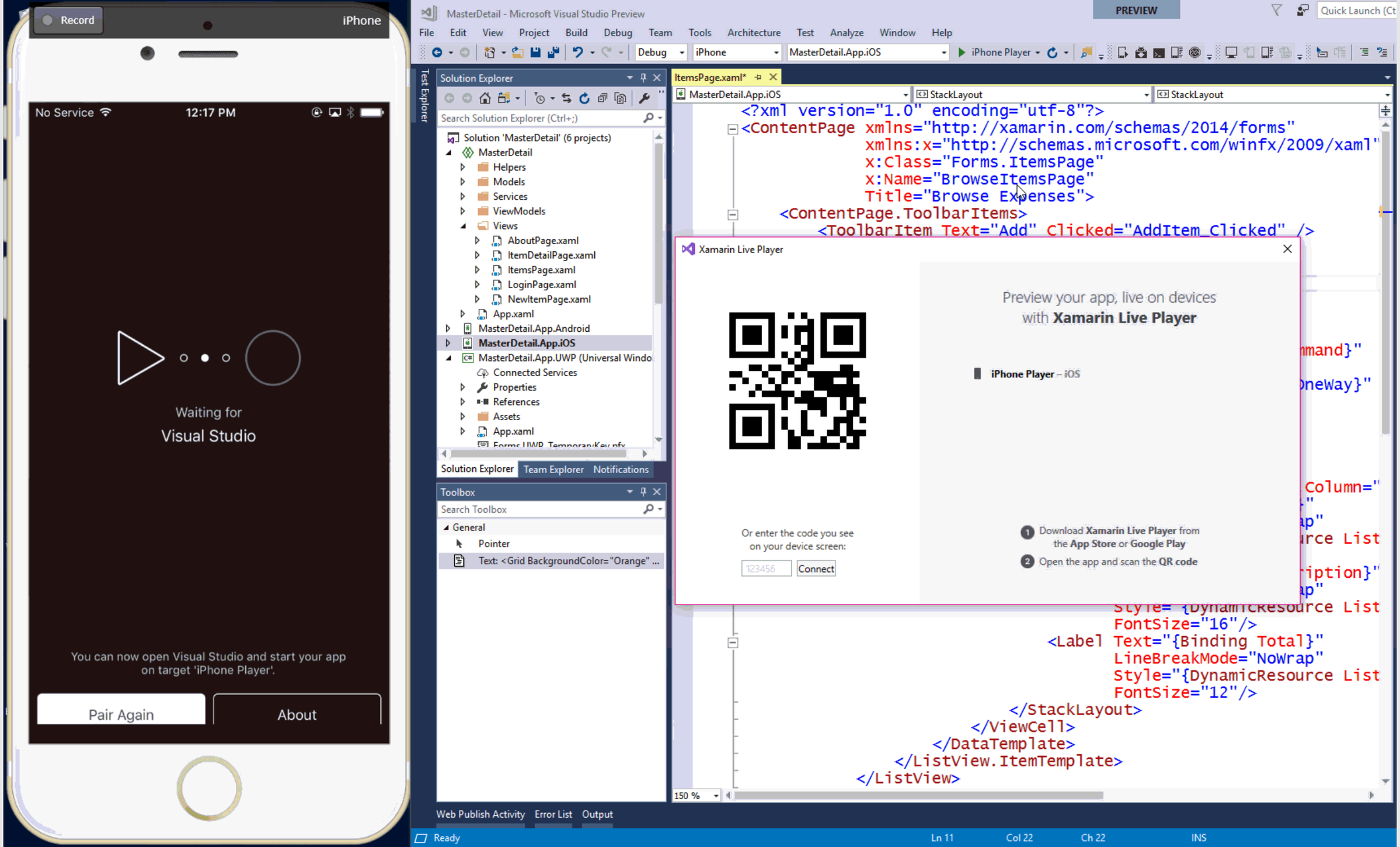
Select the Mobile development workload



Mobile Emulators

Remoted Simulator (iOS) and Android Emulators




Record
iPhone
No Service
12:17 PM
Waiting for
Visual Studio
You can now open Visual Studio and start your app
on target 'iPhone Player'.
Pair Again
About
MasterDetail - Microsoft Visual Studio Preview
PREVIEW
File Edit View Project Build Debug Team Tools Architecture Test Analyze Window Help
Debug
iPhone
MasterDetail.App.iOS
iPhone Player
Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'MasterDetail' (6 projects)
MasterDetail
Helpers
Models
Services
ViewModels
Views
AboutPage.xaml
ItemDetailPage.xaml
ItemsPage.xaml
LoginPage.xaml
NewItemPage.xaml
App.xaml
MasterDetail.App.Android
MasterDetail.App.iOS
MasterDetail.App.UWP (Universal Windo
Connected Services
Properties
References
Assets
App.xaml
Solution Explorer Team Explorer Notifications
Toolbox
Search Toolbox
General
Pointer
Text: <Grid BackgroundColor="Orange" ...
ItemsPage.xaml*
MasterDetail.App.iOS
StackLayout
StackLayout
<?xml version="1.0" encoding="utf-8"?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
x:Class="Forms.ItemsPage"
x:Name="BrowseItemsPage"
Title="Browse Expenses">
<ContentPage.ToolbarItems>
<ToolbarItem Text="Add" Clicked="AddItem_Clicked" />
Xamarin Live Player
Preview your app, live on devices
with Xamarin Live Player
iPhone Player – iOS
Or enter the code you see
on your device screen:
123456
Connect
1 Download Xamarin Live Player from
the App Store or Google Play
2 Open the app and scan the QR code
FontSize="16"/>
<Label Text="{Binding Total}"
LineBreakMode="NoWrap"
Style="{DynamicResource List
FontSize="12"/>
</StackLayout>
</ViewCell>
</DataTemplate>
</ListView.ItemTemplate>
</ListView>
150 %
Web Publish Activity Error List Output
Ready
Ln 11 Col 22 Ch 22 INS

Demo

File > New

Progressive Enhancement

Progressive enhancement

From standard experiences to total control



Styling a Forms Element

```
<Style x:Key="EntryStyle" TargetType="{x:Type Entry}">
    <Setter Property="TextColor" Value="{StaticResource WhiteColor}" />
    <Setter Property="PlaceholderColor" Value="{StaticResource WhiteColor}" />
    <Setter Property="FontSize" Value="{StaticResource LargeSize}" />
    <Setter Property="FontFamily" Value="{StaticResource KarlaRegular}" />
    <Setter Property="HorizontalOptions" Value="FillAndExpand" />
    <Setter Property="FontAttributes" Value="None" />
    <Setter Property="Opacity" Value="0.85" />
    <Setter Property="effects:LineColorEffect.ApplyLineColor" Value="True" />
    <Style.Triggers>
        <Trigger TargetType="Entry"
            Property="IsFocused"
            Value="True">
            <Setter Property="Opacity" Value="1" />
        </Trigger>
    </Style.Triggers>
</Style>
```

Styling a Forms Element

```
<Entry
      Text="{Binding Email.Value}"
      Placeholder="Email">
      <Entry.Style>
          <OnPlatform
              x:TypeArguments="Style"
              iOS="{StaticResource EntryStyle}"
              Android="{StaticResource EntryStyle}"
              WinPhone="{StaticResource UwpEntryStyle}"/>
      </Entry.Style>
</Entry>
```

Platform-Specific

```
<ContentPage
        ...
        xmlns:ios="clr-namespace:Xamarin.Forms.PlatformConfiguration.iOSSpecific;
                   assembly=Xamarin.Forms.Core”>

        <BoxView x:Name="boxView"
                 ios:VisualElement.BlurEffect="ExtraLight"
                 HeightRequest="300"
                 WidthRequest="300" />

</ContentPage>
```

Effects - XAML

```
<Label Text="Label Shadow Effect">
    <Label.Effects>
        <local:ShadowEffect Radius="5" DistanceX="5" DistanceY="5">
            <local:ShadowEffect.Color>
                <OnPlatform x:TypeArguments="Color"
                    iOS="Black"
                    Android="White"
                    WinPhone="Red" />
            </local:ShadowEffect.Color>
        </local:ShadowEffect>
    </Label.Effects>
</Label>
```

Effects - RoutingEffect

```
public class ShadowEffect : RoutingEffect
{
	public float Radius { get; set; }

	public Color Color { get; set; }

	public float DistanceX { get; set; }

	public float DistanceY { get; set; }

	public ShadowEffect () : base ("MyCompany.LabelShadowEffect")
	{
	}
}
```

Effects - PlatformEffect

```
[assembly:ResolutionGroupName ("MyCompany")]
[assembly:ExportEffect (typeof(LabelShadowEffect), "LabelShadowEffect")]
namespace EffectsDemo.iOS
{
    public class LabelShadowEffect : PlatformEffect
    {
        protected override void OnAttached ()
        {
            try {
                var effect = (ShadowEffect)Element.Effects.FirstOrDefault (e => e is ShadowEffect);
                if (effect != null) {
                        Control.Layer.CornerRadius = effect.Radius;
                        Control.Layer.ShadowColor = effect.Color.ToCGColor ();
                        Control.Layer.ShadowOffset = new CGSize (effect.DistanceX, effect.DistanceY);
                        Control.Layer.ShadowOpacity = 1.0f;
                }
            } catch (Exception ex) {
                Console.WriteLine ("Cannot set property on attached control. Error: ", ex.Message);
            }
        }
}
```

Demo

Progressive Enhancement

Xamarin.Forms Ecosystem













Awesome Xamarin Component Vendors















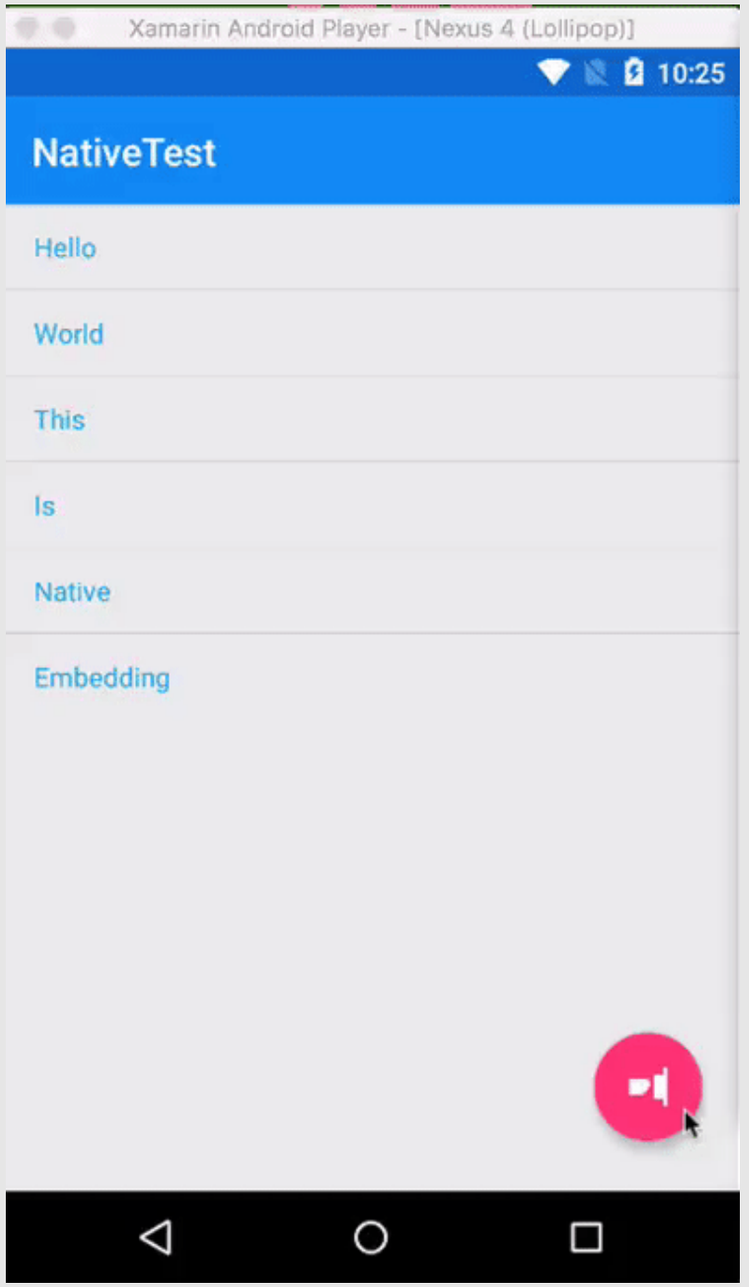




Native Control Embedding

Floating Action Button

An Android-only control that has no Xamarin.Forms abstraction.



#if to target only the desired platform

```
#if __ANDROID__
using Xamarin.Forms.Platform.Android;
using NativeTest.Droid; //Your Namespace
using Android.Views;
#endif
```

Add the FAB

```
#if __ANDROID__
var fab = new CheckableFab(Forms.Context){  UseCompatPadding = true};

fab.SetImageResource(Droid.Resource.Drawable.ic_fancy_fab_icon);

fab.Click += async (sender, e) =>{
      await Task.Delay(3000);
      await MainPage.DisplayAlert("Native FAB Clicked", "Whoa!", "OK");
};
stack.Children.Add(fab);

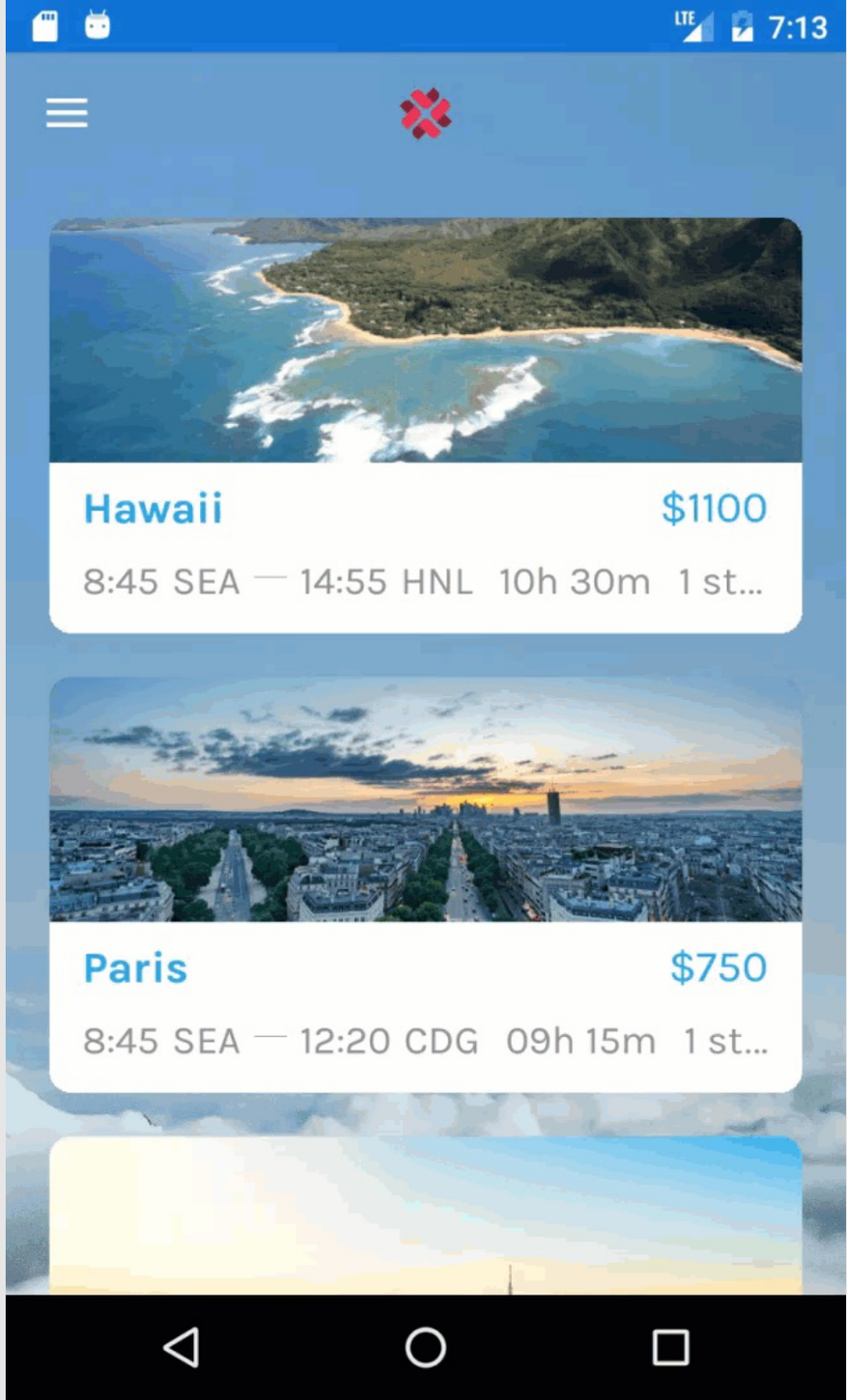
absolute.Children.Add(stack); // Overlay the FAB in the bottom-right
... // set layout and add to UI
#endif
```

Gestures

Gestures

Xamarin.Forms provides easy-to-use gestures.

- TapGestureRecognizer
- PinchGestureRecognizer
- PanGestureRecognizer



TapGestureRecognizer

```
var tapGestureRecognizer = new TapGestureRecognizer();
GestureRecognizers.Add(tapGestureRecognizer);
tapGestureRecognizer.Tapped += AnimateCardClick;

private async void AnimateCardClick(object sender, EventArgs e)
{
      var contentView = (ContentView)sender;
      var item = (Grid)contentView.Content;
      await item.ScaleTo(0.95, 50, Easing.CubicOut);
      await item.ScaleTo(1, 50, Easing.CubicIn);
}
```

What's Next for Xamarin.Forms?

2.4.0

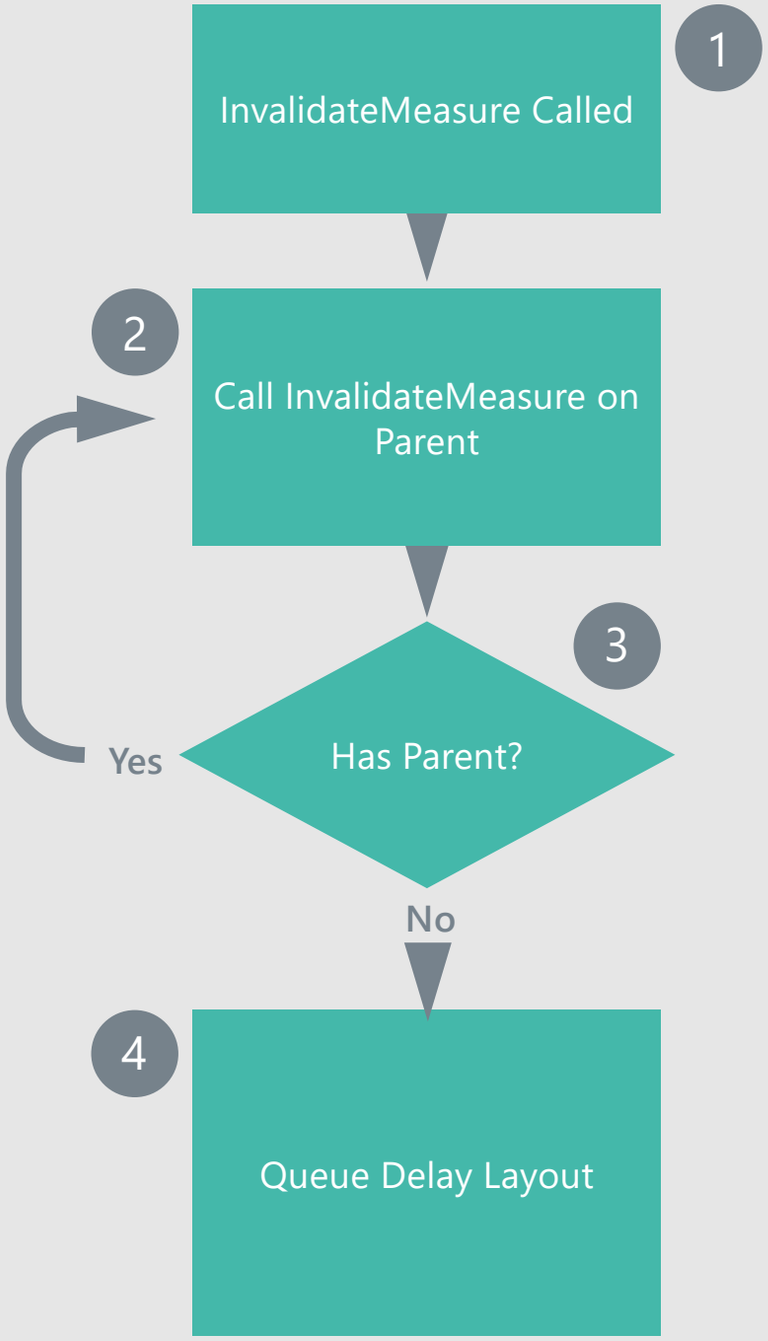
- .NET Standard 2.0
- Stability and Performance
- Over 150 fixes
- New Feature Flags API
- Android Fast Renderers: Button, Image, Label
- macOS Preview

vNext

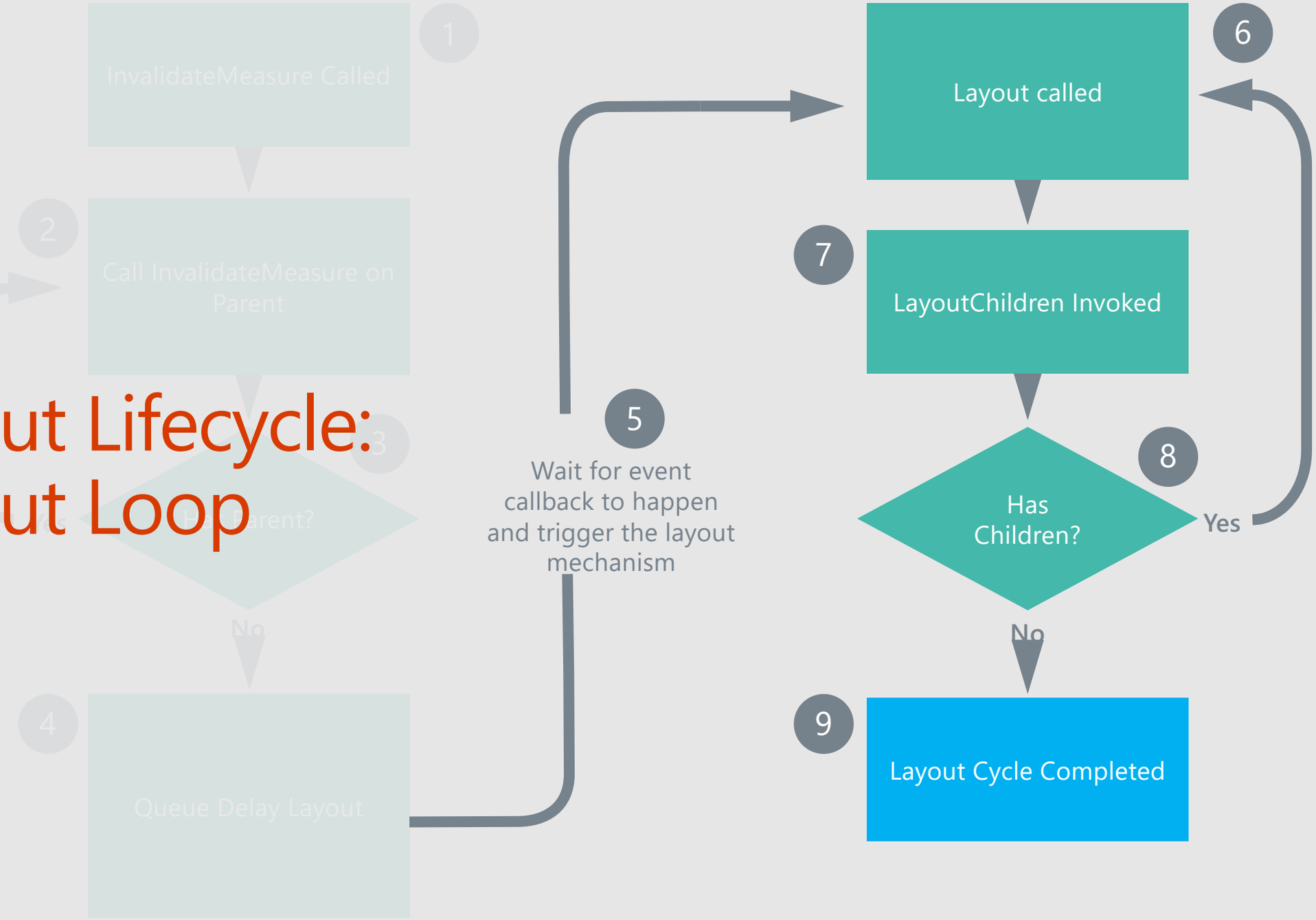
- Page Embedding
- Layout Compression
- Fast Renderers
- Right-to-Left
- ListView enhancements
- Gesture enhancements
- FlexLayout
- macOS, GTK

Layout Compression

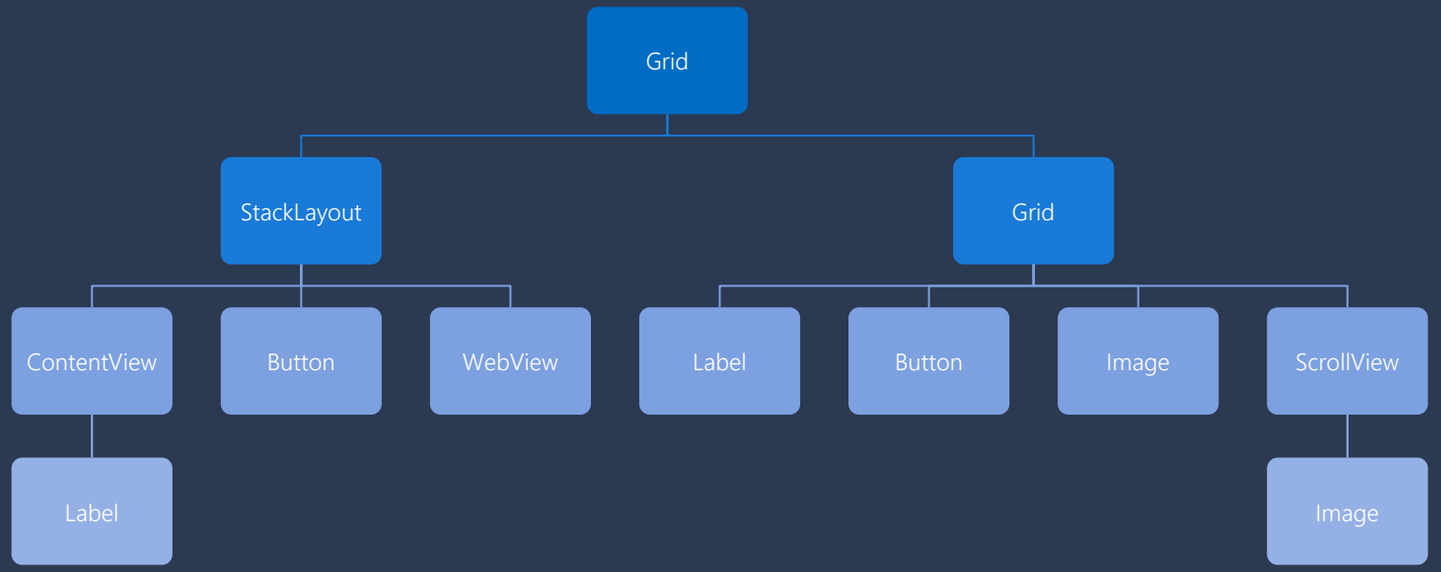
Layout Lifecycle: Invalidation Loop



Layout Lifecycle: Layout Loop

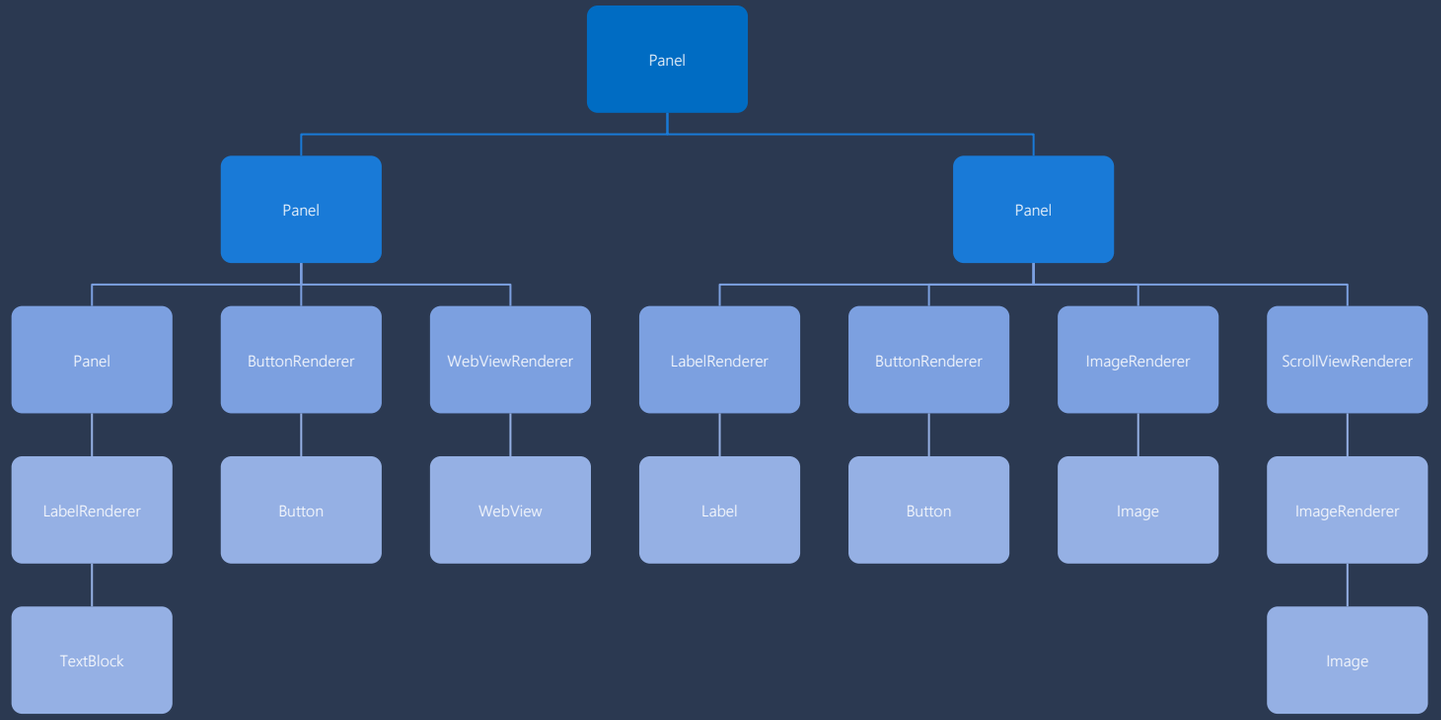


Forms



12

Platform



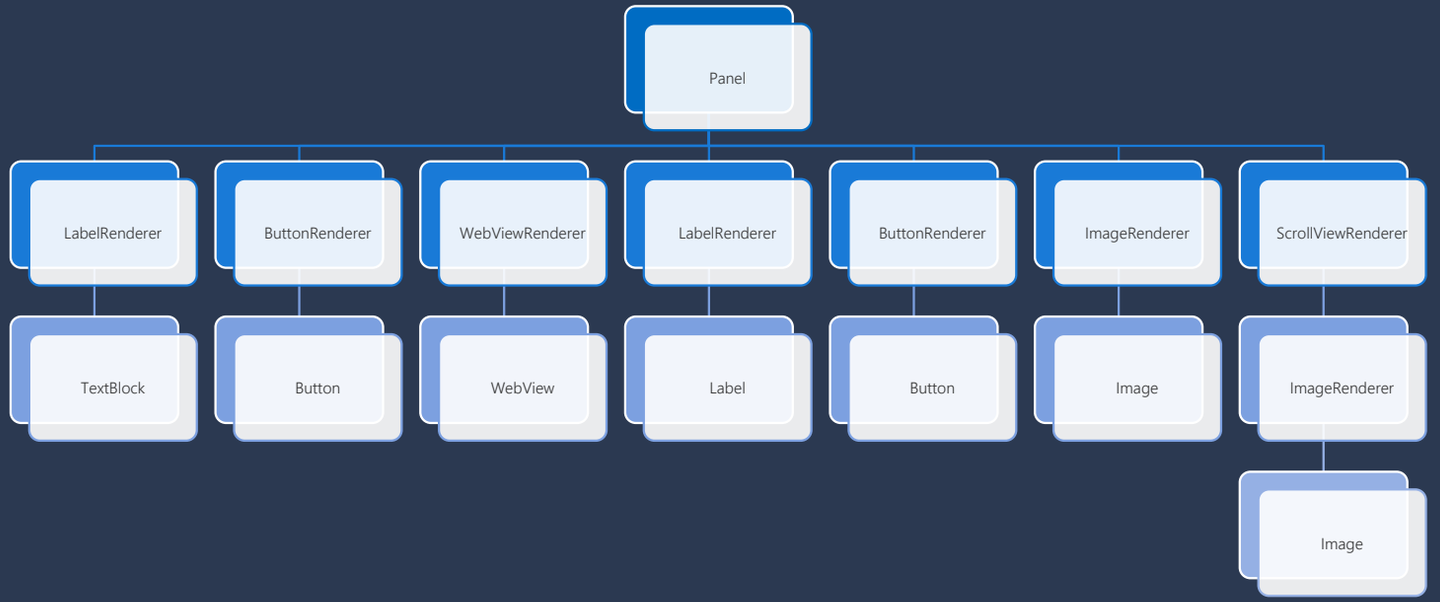
19

Platform+Compression

12

19

16



Fast Renderers

Forms

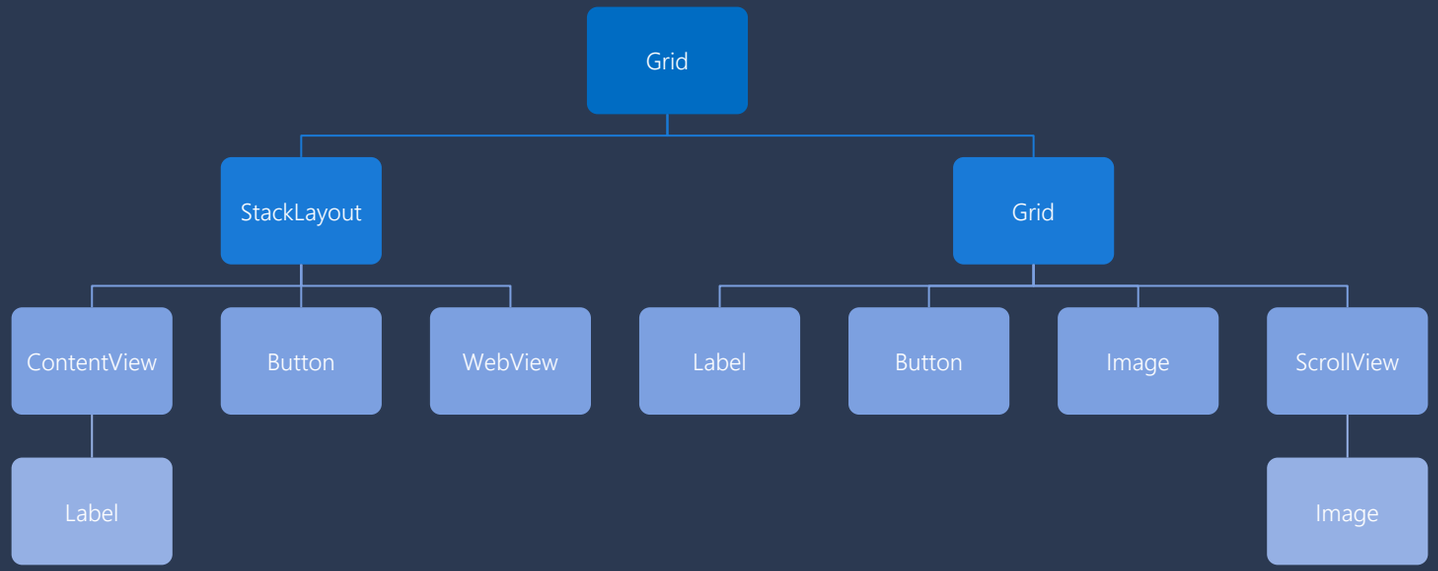
Button

Platform

ButtonRenderer

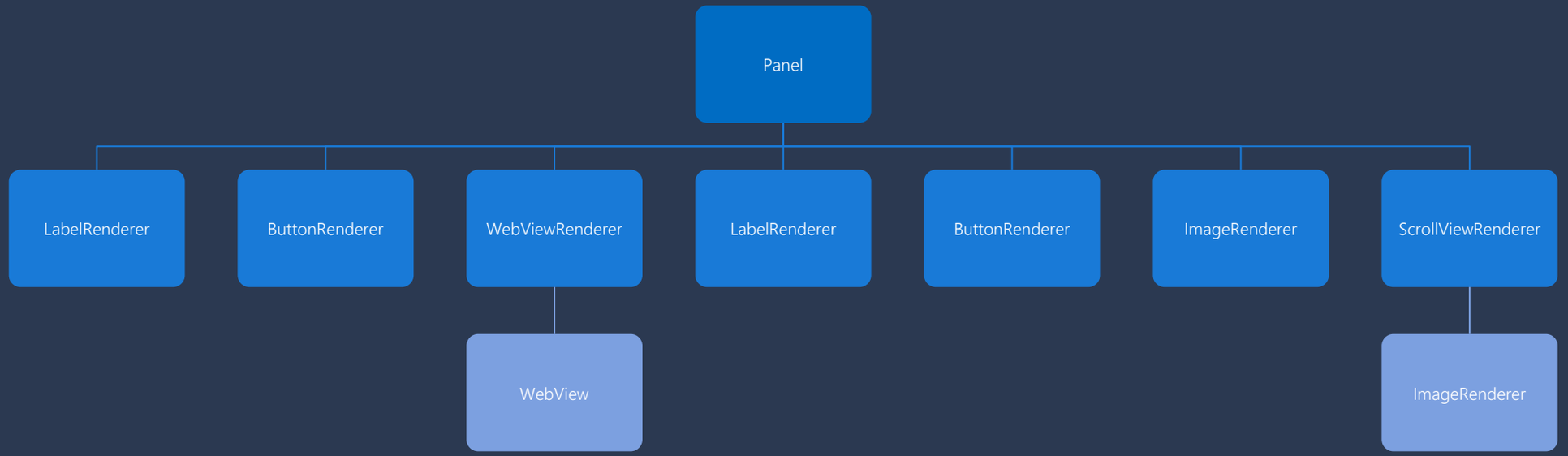
Button

Forms



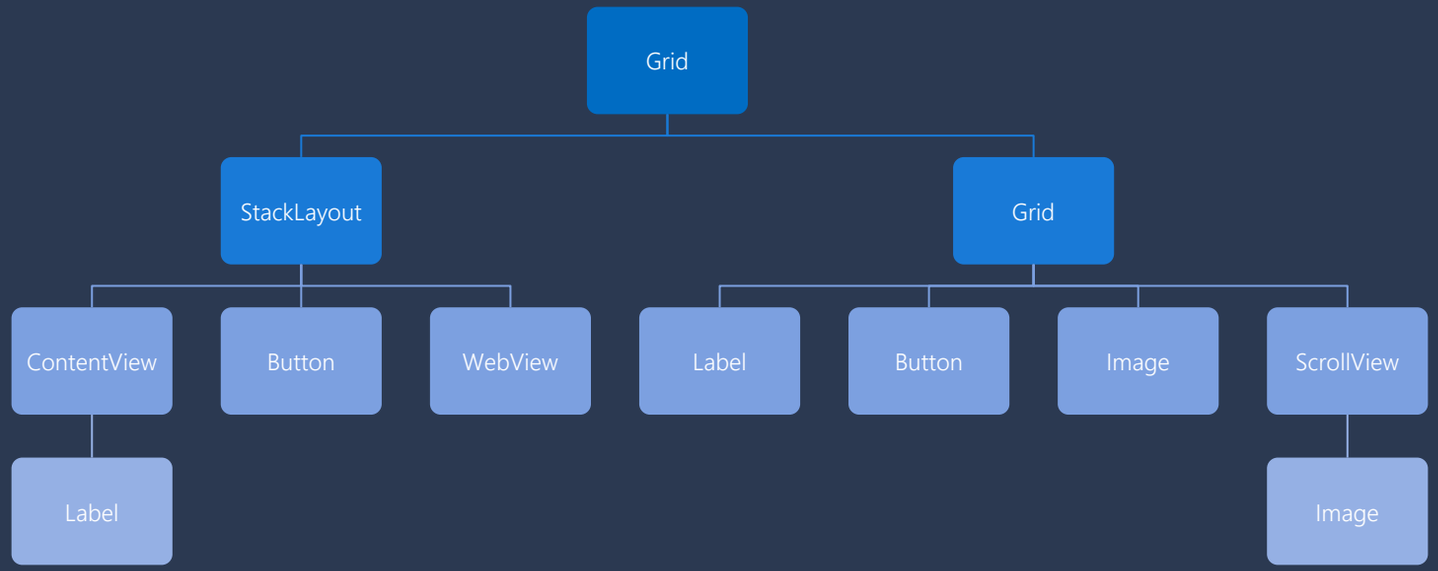
12

Platform+Compression+Fast Renderers



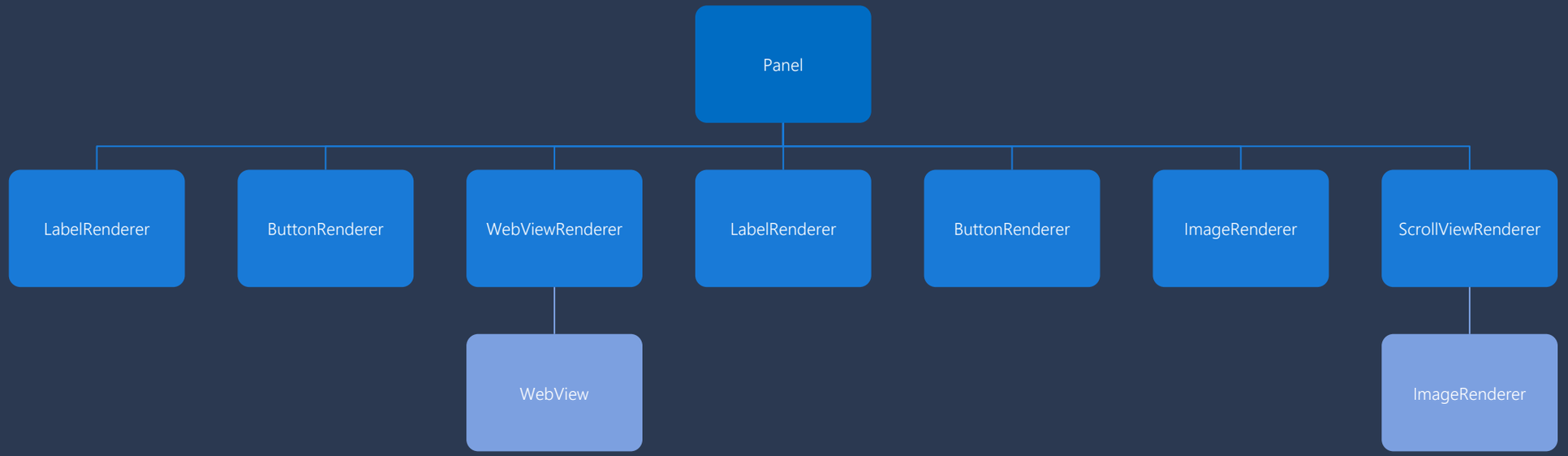
10

Forms



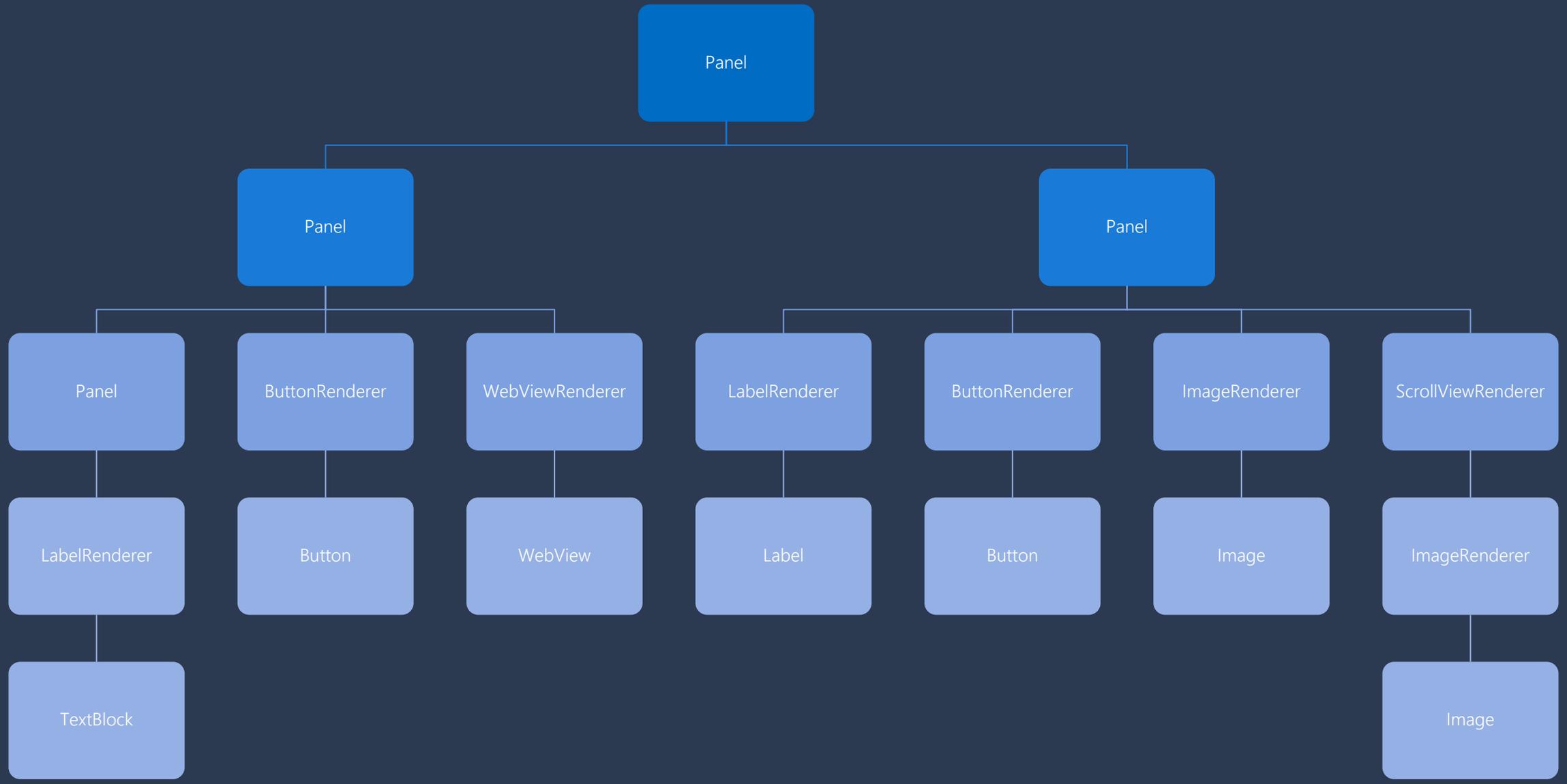
12

Platform+Compression+Fast Renderers

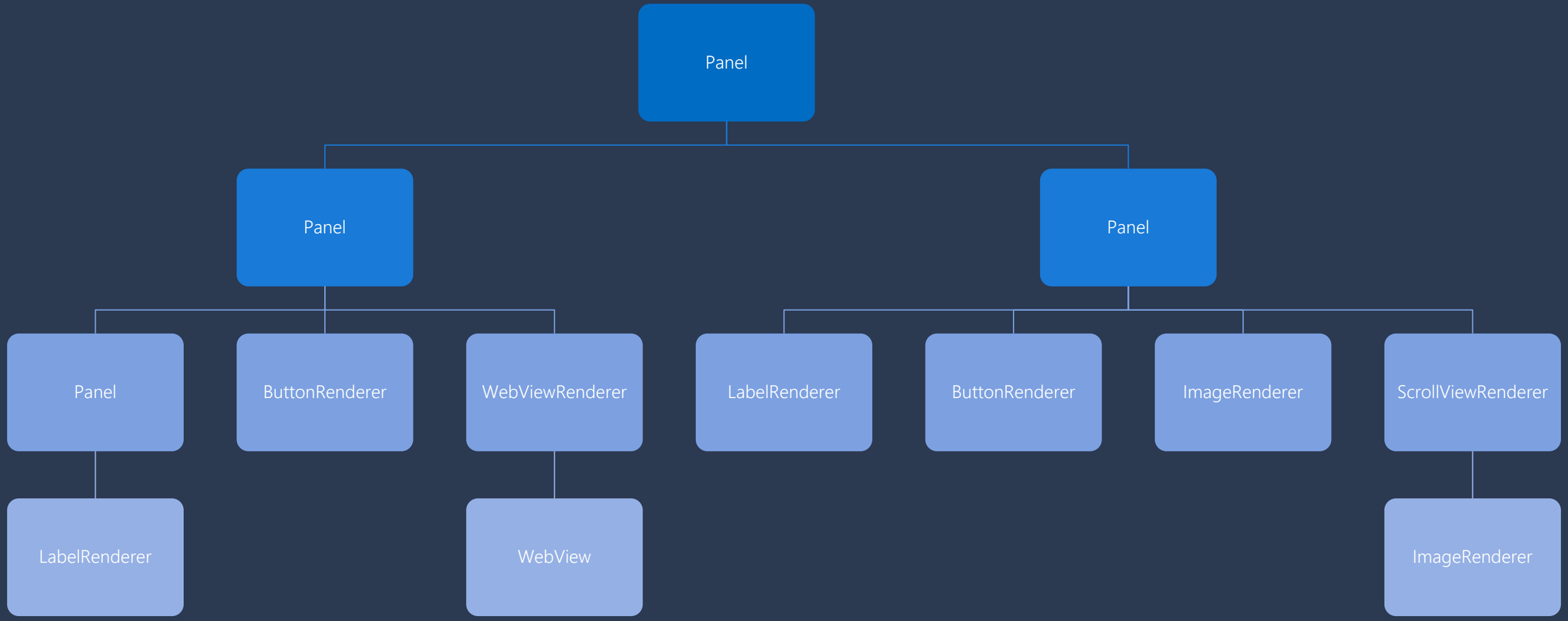


10

Forms UI Hierarchy

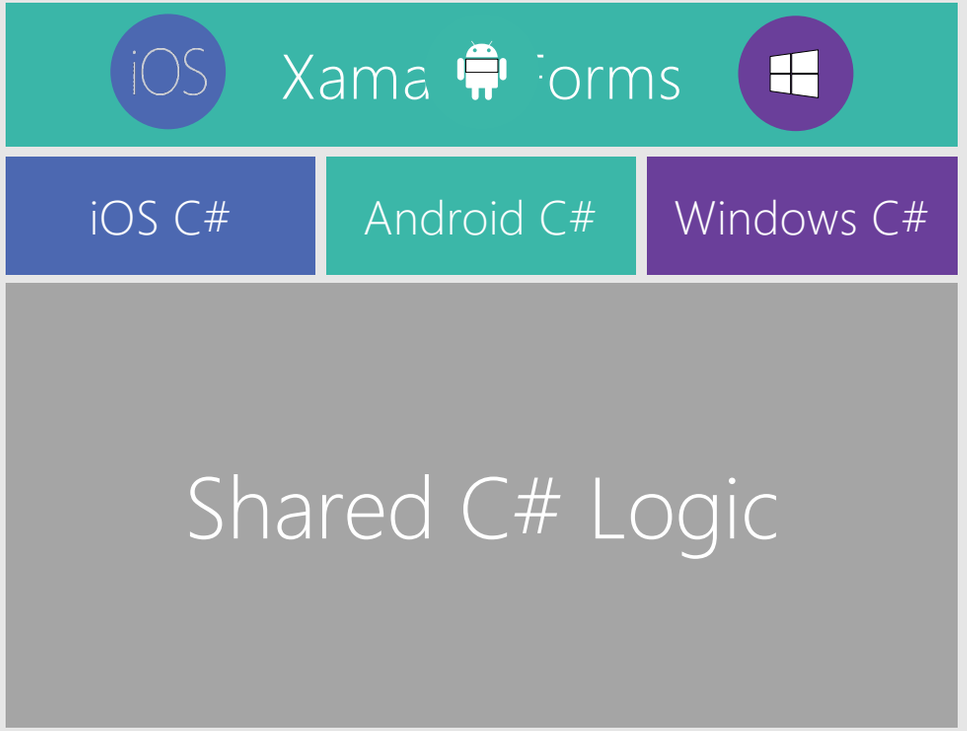


Forms UI Hierarchy with Fast Renderers



Page Embedding

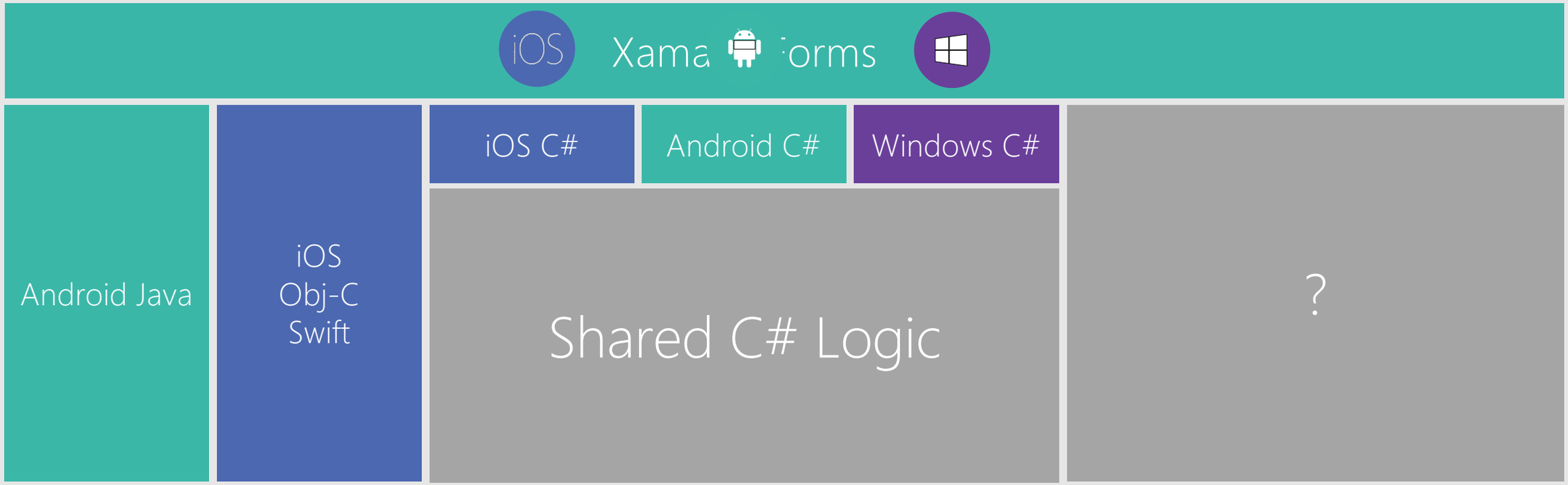
Page Embedding



Demo

Page Embedding

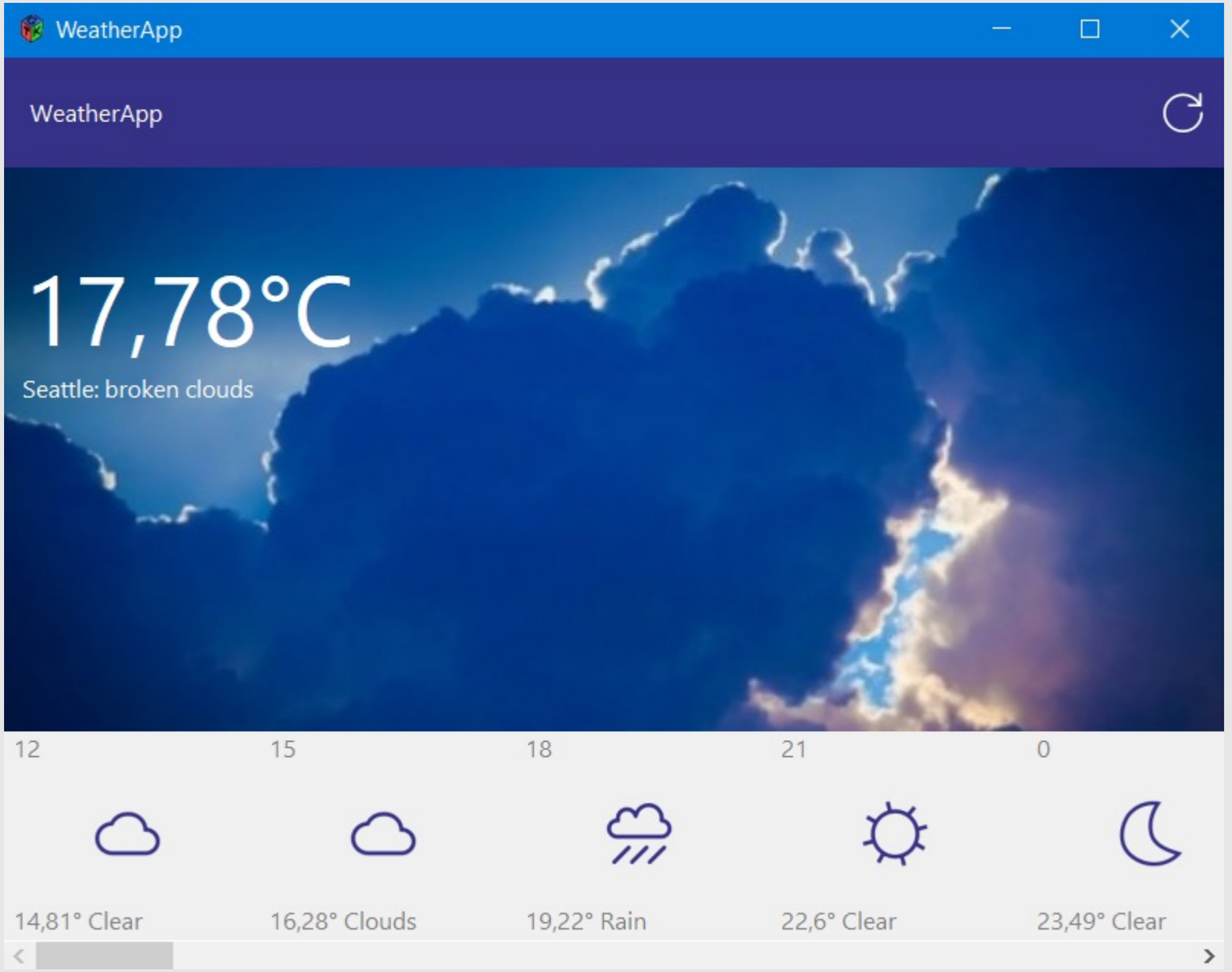
.NET Embedding



XAML Standard

- Lots of feedback!
- Reconciling the APIs we can without breakage
- Considering sharing implementations of
  - x:Bind
  - Visual State Manager
  - More

New Platforms


WeatherApp
WeatherApp
17,78°C
Seattle: broken clouds
12
15
18
21
0
14,81° Clear
16,28° Clouds
19,22° Rain
22,6° Clear
23,49° Clear


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- Follow the very latest on our Nightly Feed:
  - github.com/xamarin/Xamarin.Forms/wiki/Nightly-Builds

Microsoft

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